ZKTrustLLM-Agents L4: n8n-Orchestrated Scientific Evaluation of Proof-Governed Agentic Zero-Trust Control for O-RAN Edge Security

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***Abstract*—Autonomous O-RAN security operations require policy-governed action, auditable evidence, replay-resistant anchoring, and reproducible evaluation. This paper presents ZKTrustLLM-Agents L4, a proof-governed agentic zero-trust framework for accountable O-RAN edge security workflows. The framework decomposes autonomy into bounded telemetry, reasoning, policy, proof, and audit/anchor agents. A self-hosted n8n workflow orchestrates scientific evaluation and generates run identifiers, Git commit binding, evidence hashes, raw logs, and paper-ready KPI tables. A clean 240-record dataset is produced across six variants, four impairment profiles, and ten repetitions per condition. Stage 3 direct runtime hooks populate anchor gas, reasoning latency, replay rejection, zero-anchor rejection, and unauthorized-submitter rejection. The result is a reviewer-verifiable evaluation method for accountable autonomous O-RAN security rather than an unconstrained agentic dashboard.**

***Index Terms*—O-RAN, zero trust, agentic AI, n8n, zero-knowledge proof, blockchain audit, reproducible evaluation, KPI validation.**

I. Introduction

O-RAN disaggregates the radio access network into programmable components, including the SMO, Non-RT RIC, Near-RT RIC, rApps, and xApps. This programmability enables closed-loop optimisation but also creates security risks: unsafe autonomous action, weak evidence binding, replayed audit records, and insufficient reproducibility of security experiments.

This paper asks: *how can autonomous O-RAN security actions be made policy-gated, evidence-bound, replay-resistant, and scientifically reproducible?* We answer this through ZKTrustLLM-Agents L4, a proof-governed agentic zero-trust framework.

The novelty is not the isolated use of n8n, blockchain, IPFS, or agentic AI. The novelty is their controlled scientific composition: bounded agents reason over telemetry, policy gates restrict actions, proof paths attest admissibility, audit hooks produce direct runtime evidence, and n8n orchestrates repeatable scientific evaluation. The design is motivated by zero-trust principles [1], O-RAN policy-control workflows [2], succinct proof systems such as Groth16 [3], self-hosted command orchestration through n8n [4], and the public implementation branch used for this evaluation [5].

TABLE I
Bounded Agentic Components

Agent	Role	Output
Telemetry	Observe O-RAN/media KPIs	Jitter, loss, profile
Reasoning	Map state to trust/action	Trust state, action class
Policy	Enforce action ladder	Auto/Human/Privileged/Never
Proof	Check admissibility	AUTH_V2.x status
Audit/Anchor	Bind evidence and commit	CID/hash/anchor record

A. Contributions

The paper makes six contributions: (i) a five-agent O-RAN security architecture; (ii) a policy-gated OBSERVE–REASON–PROVE–ANCHOR–ACT loop; (iii) an n8n-orchestrated evaluation pipeline; (iv) a clean 240-record dataset; (v) Stage 3 direct runtime hooks for anchor gas, reasoning latency, replay rejection, zero-anchor rejection, and unauthorized-submitter rejection; and (vi) a fail-closed, paired public-testnet protocol for comparing identical audit-anchor bytecode on Ethereum Sepolia (PoS protocol) and IoTeX testnet (Roll-DPoS), without introducing unmeasured values into the manuscript.

II. Architecture and Threat Model

Fig. 1 shows the proposed architecture. The system places MNO policy at the SMO/Non-RT RIC layer and uses bounded agents for O-RAN security automation. The telemetry agent observes network and media-plane indicators. The reasoning agent maps state to trust level and candidate action. The policy agent enforces the action ladder. The proof agent supports bounded admissibility checks. The audit/anchor agent writes evidence hashes and compact commitments.

A. Threat Controls

The framework considers unapproved autonomous action, evidence tampering, anchor replay, zero/empty anchors, unsafe escalation, and unauthorized submitters. Fig. 2 maps these concepts to controls.

B. Smart-Contract Boundary

The smart contract does not control O-RAN traffic directly. Its role is trust-plane accountability: storing compact commitments, rejecting duplicate commitments, rejecting zero

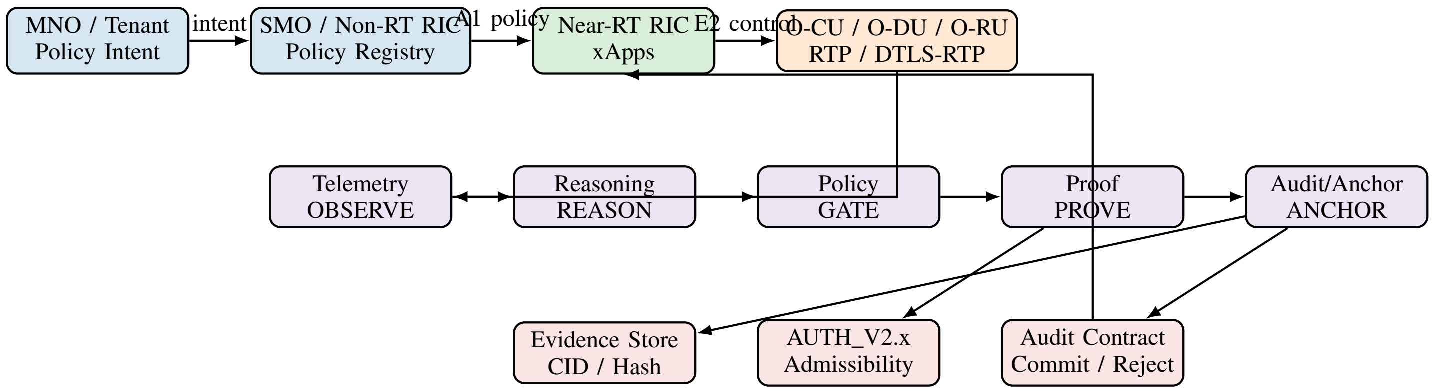

Fig. 1. Proposed ZKTrustLLM-Agents L4 architecture. Policy is placed in the SMO/Non-RT RIC governance path while bounded agents connect O-RAN telemetry to proof, evidence, and audit anchoring.

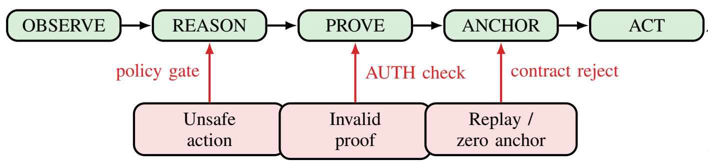

Fig. 2. Threat-to-control mapping. Unsafe actions are policy-gated, invalid proofs are blocked, and replay/zero anchors are rejected.

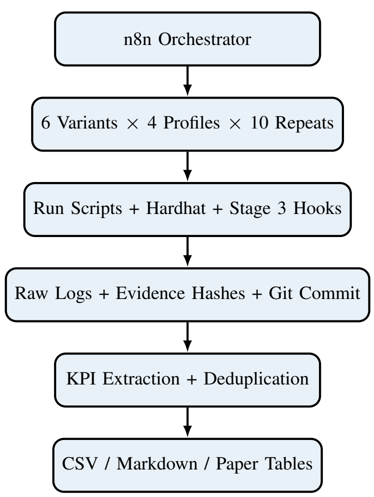

Fig. 3. n8n-orchestrated scientific evaluation pipeline. n8n coordinates execution but is not a security primitive.

commitments, and enforcing authorized submitters. It cannot verify physical network truth, guarantee evidence availability, or replace MNO operational policy.

III. Scientific Evaluation Methodology

The evaluation is organised as a three-stage pipeline. Stage 1 establishes the n8n workflow, response matrix, KPI plan, and formal-verification preparation. Stage 2 converts outputs into a clean 240-record dataset. Stage 3 adds direct runtime hooks.

A. Evaluation Matrix

The experiment covers six variants: Full L4, No-ZK, Oracle-only, RBAC-only, No-IPFS, and No-policy-gate. Each variant is evaluated over four impairment profiles and ten repetitions, giving 240 clean records.

B. KPI Classes

The dataset distinguishes direct runtime KPIs, configured impairment-profile KPIs, and missing values. This prevents overclaiming. Configured RTP/DTLS-RTP jitter and loss values are scenario-control parameters until replaced by packet-capture or O-RAN telemetry measurements.

C. Consensus-Layer Sensitivity Protocol

The consensus mechanism belongs to the execution network, not to the Solidity contract. We therefore add a paired public-testnet protocol that deploys identical `Stage3AnomalyLedger` bytecode to Ethereum Sepolia and IoTeX testnet. Sepolia follows the Ethereum PoS protocol but uses a permissioned validator set controlled by client and testing teams [6]; IoTeX documents Randomized Delegated Proof of Stake (Roll-DPoS), with delegated candidate election, VRF committee selection, and PBFT voting [7]. IoTeX testnet is EVM compatible and uses chain ID 4690 [8].

Each session executes three excluded warm-up pairs and 30 measured pairs. A matched pair carries identical commitment and evidence digests, with both transactions submitted concurrently from one client host. The publication gate requires at least three sessions, verifies chain identities and runtime-code hashes, and rejects incomplete receipts, events, block samples, or security probes. The primary metric is client-observed submission-to-first-inclusion latency; it includes RPC/client overhead and public-testnet load and is not economic finality. Median confidence intervals use a two-level hierarchical bootstrap over sessions and within-session pairs. Gas prices are recorded but not converted into a cross-asset cost comparison.

TABLE II
NOVELTY AND COMPARISON MATRIX

Approach	Agents	Policy	ZK	Evidence	Audit	Replay	KPIs
O-RAN monitoring only	No	Partial	No	No	No	No	Network only
Blockchain logging only	No	No	No	Partial	Yes	Partial	Gas/tx
RBAC contract only	No	Yes	No	No	Yes	Partial	Auth/gas
Oracle-only trust	Yes	Weak	No	Partial	Yes	No	Score/gas
No-ZK ablation	Yes	Yes	No	Yes	Yes	Yes	Workflow/gas
Full L4	Yes	Yes	Yes	Yes	Yes	Yes	Agent/ZK/audit
n8n-orchestrated L4	Yes	Yes	Optional	Yes	Yes	Yes	240 records

TABLE III
KPI COMPLETENESS AFTER STAGE 3

KPI	Coverage	Evidence Type
prover_time_ms	160/240	ablation/log gap
anchor_gas	240/240	direct runtime
rtp_jitter_ms	240/240	configured profile
rtp_loss_pct	240/240	configured profile
reason_latency_ms	240/240	direct runtime
replay_rejected	240/240	direct runtime
zero_anchor_rejected	240/240	direct runtime
unauthorized_submitter	240/240	direct runtime
postAutoScore gas	240/240	Hardhat log

IV. RESULTS

A. KPI Completeness

Table III summarises KPI completeness after Stage 3. Anchor gas, reasoning latency, replay rejection, zero-anchor rejection, and unauthorized-submitter rejection are populated from direct runtime hooks. RTP/DTLS-RTP jitter and loss are configured impairment-profile values.

B. Variant/Profile Summary

Table IV summarises the clean 240-record evaluation. The RBAC-only baseline was repaired to match the current contract ABI and rerun only for the affected variant, preserving experimental traceability.

C. Stage 3 Runtime Evidence

Stage 3 introduces an isolated Stage3AnomalyLedger micro-benchmark and a deterministic policy-reasoning probe. These emit parseable KPI lines for anchor gas, negative-security rejection, and reasoning latency.

V. DISCUSSION AND LIMITATIONS

The framework intentionally avoids overclaiming. Stage3AnomalyLedger is an isolated scientific micro-benchmark, not the production trust ledger. The deterministic reasoning probe measures bounded control-plane reasoning latency, not open-ended semantic LLM reasoning. Configured impairment-profile values are not packet-capture measurements.

The main remaining direct-runtime KPI is full ZK prover timing for full-ZK rows. Current prover-time values are populated for ablation rows where the prover is intentionally not invoked, while direct prover logs are still required for full AUTH_V2.x rows.

The PoS–Roll-DPoS extension is a consensus-sensitivity micro-benchmark rather than a claim of consensus superiority. Sepolia's validator set is permissioned, both networks are testnets, and first-inclusion latency also reflects RPC implementation, client location, propagation, and transient load. It cannot be interpreted as mainnet throughput, decentralisation, or economic-finality performance. Causal attribution beyond the directly observed latency, gas, receipt, event, rejection, and block-timestamp evidence is therefore excluded.

VI. CONCLUSION

This paper presented ZKTrustLLM-Agents L4, a proof-governed and policy-gated agentic framework for accountable O-RAN edge security workflows. The contribution is a reproducible scientific evaluation pipeline that turns an autonomous dashboard into reviewer-verifiable evidence. The clean 240-record dataset, Stage 3 runtime hooks, architecture, and KPI completeness report provide a foundation for conference and journal-level validation.

REFERENCES

[1] S. Rose, O. Borchert, S. Mitchell, and S. Connelly, "Zero trust architecture," National Institute of Standards and Technology, Tech. Rep. NIST SP 800-207, 2020.

[2] "O-ran sc a1 mediator documentation," https://docs.o-ran-sc.org/projects/o-ran-sc-ric-plt-a1/en/latest/overview.html, 2026.

[3] J. Groth, "On the size of pairing-based non-interactive arguments," in *EUROCRYPT*, 2016, pp. 305–326.

[4] "n8n execute command node documentation," https://docs.n8n.io/integrations/builtin/core-nodes/n8n-nodes-base.executecommand/, 2026.

[5] S. Karmaker, "Zktrustllm-agents l4 repository and supervisor 258 evaluation branch," https://github.com/Sudarson2021/zktrustllm-agents-l4/pull/1, 2026.

[6] Ethereum Foundation, "Ethereum networks: Sepolia," https://ethereum.org/developers/docs/networks/, accessed: 2026-07-21.

[7] IoTeX Foundation, "Randomized delegated proof of stake (roll-dpos) consensus," https://docs.iotex.io/blockchain/learn-iotex/core-concepts/consensus-mechanism, accessed: 2026-07-21.

[8] ——, "Iotex testnet and evm integration parameters," https://docs.iotex.io/blockchain/learn-iotex/exchange-integration, accessed: 2026-07-21.

TABLE IV
CLEAN 240-RECORD VARIANT/PROFILE SUMMARY

Variant	Profile	Runs	Pass	Fail	Mean Duration (ms)
full_l4	clean_baseline	10	10	0	39055.4
full_l4	delay_20ms	10	10	0	47435.9
full_l4	delay_20ms_jitter_5ms	10	10	0	56881.0
full_l4	delay_30ms_jitter_10ms_loss_1pct	10	10	0	66822.6
no_ipfs	clean_baseline	10	10	0	1782.7
no_ipfs	delay_20ms	10	10	0	1742.6
no_ipfs	delay_20ms_jitter_5ms	10	10	0	1759.8
no_ipfs	delay_30ms_jitter_10ms_loss_1pct	10	10	0	1736.5
no_policy_gate	clean_baseline	10	10	0	1741.3
no_policy_gate	delay_20ms	10	10	0	1769.1
no_policy_gate	delay_20ms_jitter_5ms	10	10	0	1739.7
no_policy_gate	delay_30ms_jitter_10ms_loss_1pct	10	10	0	1743.2
no_zk	clean_baseline	10	10	0	1765.1
no_zk	delay_20ms	10	10	0	1754.2
no_zk	delay_20ms_jitter_5ms	10	10	0	1727.6
no_zk	delay_30ms_jitter_10ms_loss_1pct	10	10	0	1789.3
oracle_only	clean_baseline	10	10	0	1777.8
oracle_only	delay_20ms	10	10	0	1752.8
oracle_only	delay_20ms_jitter_5ms	10	10	0	1783.4
oracle_only	delay_30ms_jitter_10ms_loss_1pct	10	10	0	1752.5
rbac_only	clean_baseline	10	10	0	3311.4
rbac_only	delay_20ms	10	10	0	3579.0
rbac_only	delay_20ms_jitter_5ms	10	10	0	3606.8
rbac_only	delay_30ms_jitter_10ms_loss_1pct	10	10	0	3606.6

TABLE V
STAGE 3 DIRECT RUNTIME HOOK KPIS

KPI	Coverage	Result
Anchor gas	240/240	48961 gas
Reason latency	240/240	4.889 ms
Replay rejection	240/240	true
Zero-anchor rejection	240/240	true
Unauthorized submitter	240/240	true